

DIRECT: DIrectional REasoning via grounding images with ChecklisT

Gyeom Hwangbo^{1*}, Jaemin Kim^{2*}, and Yumin Seol^{1*}

¹ Pohang University of Science and Technology, South Korea

² Korea Advanced Institute of Science and Technology, South Korea

gyeomhwangbo@postech.ac.kr, jaemkim01@kaist.ac.kr, seolym@postech.ac.kr

Abstract. The CoIN Challenge 2026 evaluates multimodal agents that determine whether each sequentially presented candidate image matches a fixed target, prioritizing identification accuracy while minimizing clarification questions. Success hinges on two coupled decisions: *what to ask* for discriminative evidence and *when to stop asking* without compromising reliability. Among many plausible questions, only a few provide evidence that distinguishes visually similar candidates, while the evidence accumulated from those questions determines whether further clarification is necessary. We propose DIRECT (**DI**rectional **RE**asoning via grounding images with **C**hecklis**T**s), a stateful reasoning strategy that addresses both decisions jointly. DIRECT grounds each candidate by category and organizes clarification around directional zones to elicit spatially grounded, discriminative evidence. It stores the resulting evidence as verified claims in an append-only checklist, enabling reuse across candidates, rejection upon clear contradictions, and termination once sufficient evidence has accumulated. With deterministic checklist updates and conservative contradiction tests, DIRECT supports reliable target identification while reducing redundant interaction. The code is publicly available at <https://github.com/gyeomh/DIRECT>

1 Introduction

Collaborative Instance-object Navigation (CoIN) studies embodied agents that must identify and navigate to a target object instance specified through natural language in cluttered environments containing visually similar distractors [1]. When the available description and visual observation are ambiguous, a capable agent should actively request clarification instead of committing to an unreliable prediction. The CoIN Challenge 2026 requires multimodal agents to determine whether each sequentially presented candidate image matches a fixed target, asking clarifying questions when the available information is ambiguous. It evaluates both identification accuracy and the number of clarification questions asked, placing greater emphasis on accuracy while using a vision-language model as a reproducible user oracle [2].

*Equal contribution.

Successful target identification in this challenge hinges on two coupled sub-problems: *what to ask* and *when to stop asking*. Although an agent can formulate many plausible questions about an image, only a small subset yields evidence that meaningfully distinguishes the target from visually similar distractors, which often share category-level appearance and differ only in subtle attributes or spatial context. The agent must therefore determine which specific regions and attributes to ask about. It must also determine when the accumulated evidence is sufficient to accept or reject a candidate: stopping too early risks an incorrect decision, whereas continuing after the relevant ambiguity has been resolved incurs unnecessary interaction. Moreover, because the oracle consistently describes the same fixed target throughout an episode, treating each candidate independently can lead the agent to revisit previously resolved ambiguities, wasting information acquired from earlier interactions. These two decisions are inherently coupled: the questions selected by the agent determine what evidence is accumulated, while the accumulated evidence determines whether further clarification is necessary.

To address both decisions jointly, we propose **DIRECT** (**DI**rectional **RE**asoning via grounding images with **C**hecklis**T**s), a stateful reasoning strategy that organizes clarification around directional zones and maintains an append-only checklist of verified claims about the target. Directional question generation focuses each interaction on spatially grounded, discriminative evidence, thereby structuring what to ask, while the checklist preserves this evidence across candidates and enables the agent to stop asking once sufficient evidence has been accumulated for a reliable decision.

Combined with category-only grounding, deterministic checklist updates, and conservative contradiction tests, the proposed strategy is intended to reduce redundant dialogue and mitigate false rejection of the true target under ambiguity. This design therefore supports the challenge objective of maximizing instance-identification accuracy while minimizing user questions.

2 DIRECT: Directional Checklist Verification

We propose DIRECT, a questioner that maintains a *checklist* of assertions about the true target and interrogates each candidate through *directional regions* anchored to that target.

2.1 Overview

DIRECT takes as input a natural-language description d and a sequence of candidate images $\langle I_k \rangle_{k=1}^K$, exactly one of which corresponds to the target. It may interact with an oracle $\mathcal{O}$, a vision-language model that has access only to the target image and answers free-form questions about it. Candidates are presented sequentially, and DIRECT must reach a decision for each candidate before the next candidate is revealed. For candidate I_k , DIRECT uses the accumulated checklist C_{k-1} to verify the candidate, queries $\mathcal{O}$ about a set of directional regions, and

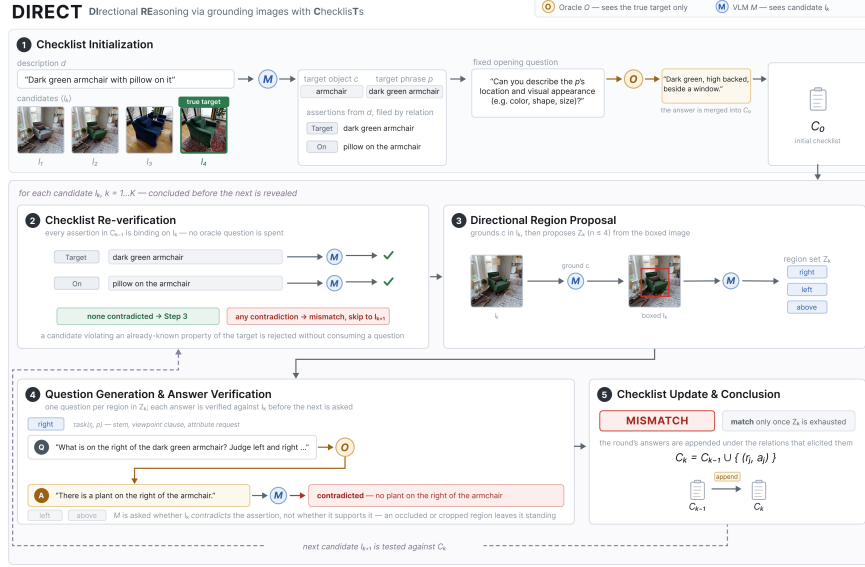


Fig. 1: Overview of our proposed method.

produces a binary decision $\hat{y}_k \in \text{MATCH}, \text{MISMATCH}$ together with an updated checklist C_k :

$$\hat{y}_k, C_k \leftarrow \text{DIRECT}(I_k \mid C_{k-1}, \mathcal{O}). \quad (1)$$

As illustrated in Figure 1, our framework decomposes process into five steps.

2.2 Step 1: Checklist Initialization.

The natural-language description d may specify multiple requirements that simultaneously hold for the target, each of which must be verified independently for every candidate. A VLM M therefore decomposes d into an initial checklist $C_0 = (r_i, a_i)_{i=1}^m$, where a_i is a natural-language assertion and $r_i \in \mathcal{R}$ specifies the spatial relation under which the assertion is verified. Since assertions associated with different regions require evidence from different parts of an image, each assertion is assigned to one of 11 relation symbols:

- **Cardinal**: left, right, above, below.
- **Diagonal**: the four compound relations, e.g., left-above.
- **On**: objects resting on the target’s top surface.
- **Next to**: objects whose relative side is unspecified or described only by a depth cue, such as “in front of,” which cannot be represented by $\mathcal{R}$.
- **Target**: attributes of the target itself, with no spatial reference.

In addition, M identifies the target object c and the target phrase p . The former is used for grounding, while the latter, which also encodes the target’s attributes,

is used to formulate oracle queries. Because the description may leave important aspects of the target’s appearance unspecified, DIRECT begins each episode by issuing a single fixed query to the oracle: “*Can you describe the p ’s location and visual appearance (e.g., color, shape, size)?*” The oracle’s response is then incorporated into C_0 . This is the only query in the framework that is not conditioned on a specific relation.

2.3 Step 2: Checklist Re-verification.

Each assertion in C_{k-1} is elicited from $\mathcal{O}$, which observes only the true target, and therefore describes the target rather than the candidate from which it is obtained. Once established, an assertion is thus valid for all subsequent candidates. Before querying $\mathcal{O}$ about I_k , DIRECT first verifies every assertion in C_{k-1} . Specifically, for each $(r_i, a_i) \in C_{k-1}$, M produces a verdict v_i and a short evidence span e_i indicating whether I_k contradicts a_i within the region $\tau_{\text{reg}}(r_i, c)$:

$$v_i, e_i \leftarrow M(I_k, \tau_{\text{reg}}(r_i, c), a_i). \quad (2)$$

If any v_i indicates a contradiction, DIRECT immediately predicts $\hat{y}_k = \text{MISMATCH}$ and proceeds to I_{k+1} . This oracle-free verification step therefore rejects candidates that violate any previously established target property without consuming an additional oracle query.

2.4 Step 3: Directional Region Proposal.

For a candidate that survives Step 2, DIRECT generates a set of directional regions to formulate questions for further verification. First, M grounds the target in I_k using the target object c , and overlays the resulting bounding box on the image. Conditioned on the boxed image, M proposes a region set $Z_k = \{r_j\}_{j=1}^n$ with $n \leq 4$.

We then apply two filters to Z_k . First, we remove any region that already has assertions in C_{k-1} , ensuring that each direction is queried at most once per episode. Second, we remove any region that falls outside the image boundary, as its contents cannot be reliably verified. If both filters remove all regions, we fall back to the cardinal directions that remain within the image boundary, ensuring that every candidate is evaluated on at least one region.

2.5 Step 4: Question Generation and Answer Verification.

Each region in Z_k generates one oracle query, whose answer is verified against the candidate before proceeding to the next region. For each $r_j \in Z_k$, a shared template maps the relation to both a query $\tau_{\text{ask}}(r_j, p)$ and a corresponding image region $\tau_{\text{reg}}(r_j, c)$, ensuring consistent relation semantics across the oracle and candidate paths:

$$a_j \leftarrow \mathcal{O}(\tau_{\text{ask}}(r_j, p)), \quad v_j, e_j \leftarrow M(I_k, \tau_{\text{reg}}(r_j, c), a_j). \quad (3)$$

Each query consists of a relation-specific stem and a shared request for visual attributes, with an additional viewpoint clause for lateral relations. For example, for $r_j = \text{left}$:

“What is on the left of the p? Judge left and right as you see them looking at the image, not from the p’s own point of view. Can you describe the shape and color?”

The viewpoint clause is omitted for vertical relations, while the **on** relation explicitly refers to the target’s top surface.

As in Step 2, a contradiction immediately yields $\hat{y}_k = \text{MISMATCH}$, and the remaining regions are not queried. Importantly, M determines whether the candidate contradicts the oracle’s assertion rather than whether it supports it. Thus, ambiguous evidence caused by occlusion, cropping, or insufficient resolution does not trigger rejection. This asymmetric criterion reduces false rejections of the target, while remaining safe because distractors can still be rejected by subsequent assertions. Each assertion is verified independently, so a single verdict never covers multiple claims.

For color attributes, M reports the perceived color in e_j , while DIRECT determines color-family equivalence using a predefined table indexed by hue and a three-level lightness scale. A perceived color therefore overrides a contradiction when the claimed and perceived terms belong to the same family (e.g., *light brown* and *beige*), while terms from different families remain contradictory. This one-directional override prevents minor color-label variations from causing false rejection without allowing the normalization to introduce a new rejection.

2.6 Step 5: Checklist Update and Conclusion.

DIRECT predicts $\hat{y}_k = \text{MATCH}$ only when all regions in Z_k have been queried and their answers verified against I_k without contradiction. Thus, termination is determined by the exhaustion of the geometrically admissible and informative regions in Z_k , rather than by a confidence threshold or a fixed query budget.

The verified answers are then appended to the checklist under their corresponding relations:

$$C_k = C_{k-1} \cup (r_j, a_j)_{j=1}^n. \quad (4)$$

The checklist grows monotonically: assertions retain their original wording and are never edited, reordered, or removed. Consequently, any assertion acquired while processing I_i remains available for verifying every subsequent candidate I_j , $j > i$. Each oracle answer is appended as a single assertion rather than decomposed into atomic claims, preserving modifiers and their associated nouns.

3 Results

3.1 Experimental Setup

Dataset. We evaluate on the ECCV 2026 CoIN Challenge benchmark. Each episode provides a natural-language description of a target object together with a

sequence of candidate images, exactly one of which is the true target; the number of candidates per episode ranges from 1 to 7 with a mean of 3.16. Every episode is annotated with one of six *description types*, which differ in the kind of information the description carries: **category** (a bare category noun), **color**, **context** (relational wording about surrounding objects), **color_feature**, **color_context**, and **color_context_feature**. The evaluation set contains 167 episodes per type, 1002 episodes in total.

Metrics. An episode counts as solved only if the conclusion is correct for *every* candidate in it, so accuracy is all-or-nothing per episode; the challenge score of a solved episode is $10 - n_q$, where n_q is the number of oracle questions consumed. Each episode is capped at 60 environment steps and 600 seconds of wall-clock time, shared across all of its candidates. Beyond accuracy, the challenge scores a questioner in *points*, which additionally charge for the evidence a conclusion was reached with. An episode concluded correctly is awarded +10 and an episode concluded incorrectly −10, and each oracle question consumed costs a further −1 regardless of the outcome, so that a single episode contributes

$$s = \begin{cases} +10 - n_q & \text{if the episode is solved,} \\ -10 - n_q & \text{otherwise,} \end{cases} \quad (5)$$

and the reported total is $S = \sum_e s_e$ over all 1002 episodes.

Discarded episodes. 56 of the 1002 episodes are discarded by the harness before any questioner acts on them, owing to a data-side issue in which two consecutive candidate images are byte-identical; the harness treats this as an unrecoverable error irrespective of the conclusions submitted. These discards are a property of the data rather than of any questioner’s design and fall almost evenly across types (9–10 each), so we report every metric under two conventions: *all episodes*, which follows the benchmark’s own convention and counts each discard as a failure charged −10, and *valid episodes*, which restricts the denominator to the 946 episodes actually logged.

Model. All modules of DIRECT are instantiated with a single vision–language model, Qwen3.6-35B-A3B-FP8, served with vLLM. We use temperature = 0.0 throughout and disable the model’s reasoning blocks. Every module’s output is pinned by strict JSON-schema guided decoding, and prefix caching is enabled. The oracle $\mathcal{O}$ is supplied by the benchmark harness and observes the true target image alone; DIRECT interacts with it only through free-form questions. No component is trained or fine-tuned: DIRECT is a prompting-and-control architecture over a frozen model.

3.2 Quantitative Results

Table 1 reports the points DIRECT earns under Eq. (5) for each description type and over the full benchmark. It also reports the number of solved episodes and the

Table 1: Results of DIRECT on the Coin Challenge benchmark, per description type (167 episodes per type) and overall. *Solved* counts episodes in which every candidate is concluded correctly, so accuracy is all-or-nothing per episode; $\bar{n}_q$ is the mean number of oracle questions per episode; *Points* follows the challenge scoring of Eq. (5).

Description type	All episodes				Valid episodes			
	Solved	Acc. (%)	$\bar{n}_q$	Points	Solved	Acc. (%)	$\bar{n}_q$	Points
category	139 / 167	83.2	3.78	478	139 / 158	88.0	4.00	568
color	138 / 167	82.6	3.64	482	138 / 158	87.3	3.85	572
context	140 / 167	83.8	3.34	573	140 / 158	88.6	3.53	663
color_feature	129 / 167	77.2	3.39	344	129 / 158	81.6	3.58	434
color_context	139 / 167	83.2	3.31	557	139 / 157	88.5	3.52	657
color_context_feature	129 / 167	77.2	2.92	423	129 / 157	82.2	3.10	523
Overall	814 / 1002	81.2	3.40	2857	814 / 946	86.0	3.60	3417

mean number of questions per episode. DIRECT solves 814 of the 1002 episodes. This corresponds to an accuracy of 81.2% under the benchmark’s official convention and 86.0% over the episodes logged by the harness. This accuracy comes from verifying the target and its surrounding directional relations. DIRECT uses multiple spatial cues to distinguish the true target from distractors rather than relying on a specific region of the image.

DIRECT achieves this performance with few oracle questions. It asks an average of 3.40–3.60 questions per episode. No description type requires more than 4.00 questions per episode. Thus, a correctly solved episode retains at least 6 points on average after the question penalty. The method therefore operates well within the scoring margin. This efficiency comes from the combination of directional questioning and checklist reuse. Directional regions restrict each query to a specific spatial relation, while the checklist tracks which target properties have already been verified. Once all required properties are verified without contradiction, DIRECT concludes MATCH without further queries.

In result, DIRECT earns 2857 points under the official convention and 3417 points over the valid episodes. The difference is exactly the -10 penalty applied to each of the 56 discarded episodes.

4 Conclusion

Interactive instance identification requires an agent to decide both which regions and attributes to ask about, given that visually similar distractors often differ only in subtle attributes or spatial context, and when the accumulated evidence suffices to accept or reject a candidate. We addressed these coupled decisions with DIRECT, which grounds the target category in each candidate, queries directional regions around it, and verifies answers under a conservative contradiction criterion. Verified assertions persist in a checklist, so evidence acquired earlier is reused rather than re-elicited, and termination follows from exhausting the admissible regions rather than a fixed budget. DIRECT attains strong accuracy with few clarifying questions, and the checklist proves particularly effective

as an episode progresses, since assertions established on early candidates reject later distractors without further interaction.

Future work. Accuracy holds consistently across description types except for feature descriptions, which characterize the target’s own appearance and thus resist localization to a directional region, suggesting that complementing directional reasoning with target-centric verification could be a promising direction to explore.

References

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